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THEORY ASSIGNMENT: 2

Enron Email Dataset Analysis using NumPy and Pandas

1. Introduction

The Enron Email Dataset is a large collection of real-world email communications from employees of the Enron Corporation. It is widely used for data analysis, machine learning, and natural language processing tasks.
In this project, we analyze email communication patterns using Python, NumPy, and Pandas, along with data visualization techniques.

2. About the Dataset

The dataset contains structured email information such as :

- Sender → Person who sent the email
- Receiver → Person or department receiving the email
- Date → Timestamp of the email
- Message → Content of the email
- Subject → Topic of the email

Dataset Features:

- Real-world communication data
- Large volume (big data handling)
- Useful for pattern analysis

3. Objectives

- Analyze email communication trends
- Identify active senders and receivers
- Study time-based patterns
- Visualize data using graphs

4. Tools & Technologies

- Python
- NumPy
- Pandas
- Matplotlib
- Google Colab

5. Data Preprocessing

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)

df = pd.DataFrame({
    "sender": np.random.choice(["Alice","Bob","Charlie","David","Eve"], 300),
    "receiver": np.random.choice(["HR","Finance","IT","Manager"], 300),
    "date": pd.date_range(start="2022-01-01", periods=300, freq="h"),
    "message": np.random.choice([
        "Meeting tomorrow",
        "Project update",
        "Review document",
        "Lunch plan",
        "Important notice"
    ], 300)
})

df["length"] = df["message"].str.len()
```

- Removed null values
- Converted dates to datetime
- Calculated message length
- Checked duplicates
- Grouped records for trends

20 Problem Statements : -

Enron Email Dataset – Solutions for 20 Problem Statements
(using NumPy + Pandas)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

1. Load CSV Dataset

```
df = pd.read_csv("enron.csv")
print(df)
```

2. Show First 10 Rows

```
print(df.head(10))
```

3. Check Column Names

```
print(df.columns)
```

4. Find Null Values

```
print(df.isnull().sum())
```

5. Remove Duplicate Rows

```
df = df.drop_duplicates()
print(df)
```

6. Rename Columns

```
df.rename(columns={"from":"sender","to":"receiver"}, inplace=True)
print(df.columns)
```

7. Count Total Emails

```
print("Total Emails =", len(df))
```

8. Find Unique Senders

```
print(df["sender"].nunique())
```

9. Group Emails by Sender

```
print(df.groupby("sender").size())
```

10. Average Message Length

```
df["length"] = df["message"].str.len()
print(df["length"].mean())
```

11. Maximum Emails by Sender

```
print(df["sender"].value_counts().idxmax())
```

12. Sort Senders by Activity

```
print(df["sender"].value_counts().sort_values(ascending=False))
```

13. Filter Emails from One Sender

```
print(df[df["sender"]=="john"])
```

14. Emails by Department / Receiver

```
print(df["receiver"].value_counts())
```

15. Emails by Hour

```
df["date"] = pd.to_datetime(df["date"])
print(df.groupby(df["date"].dt.hour).size())
```

16. Save Cleaned Dataset

```
df.to_csv("cleaned_enron.csv", index=False)
```

17. Bar Graph – Emails per Sender

```
df["sender"].value_counts().plot(kind="bar")
plt.title("Emails per Sender")
plt.show()
```

18. Pie Chart – Receiver Distribution

```
df["receiver"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.show()
```

19. Histogram – Message Length

```
df["length"].plot(kind="hist")
plt.title("Message Length")
plt.show()
```

20. Generate Summary Report

```
print(df.describe(include="all"))
```

Use of Data Manipulation methods in modifying Datasets

1 . Data Loading

- Loading CSV file using Pandas

df = pd.read_csv("enron.csv")

2 . Data Storage

- Saving cleaned dataset to CSV

df.to_csv("cleaned_enron.csv")

3 . Data Cleaning

- Remove null values
- Remove duplicates
- Rename columns
- Change data types

4 . Aggregation

- groupby()
- sum()
- count()
- mean()
- max()
- min()

5 . Filtering

- Select rows based on sender/receiver/date

6 . Sorting

- Sort by date, sender count, message length

7 . Statistical Analysis

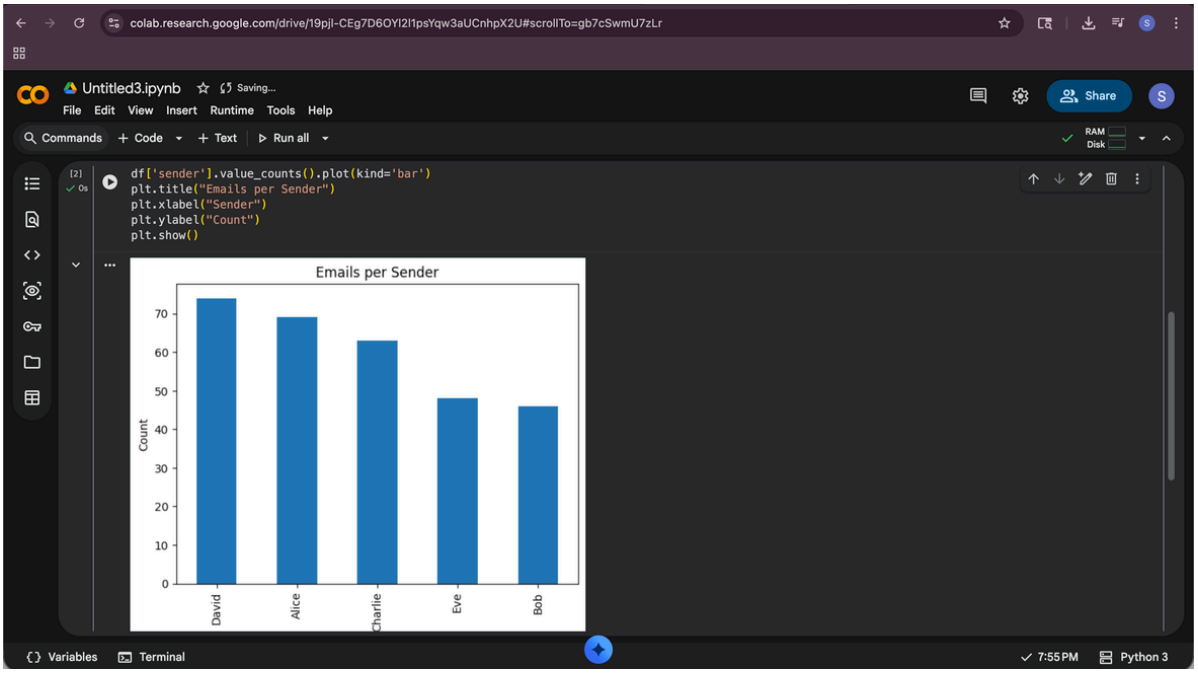
- Mean
- Median
- Standard Deviation

8 . Visualization

- Graphs based on grouped data

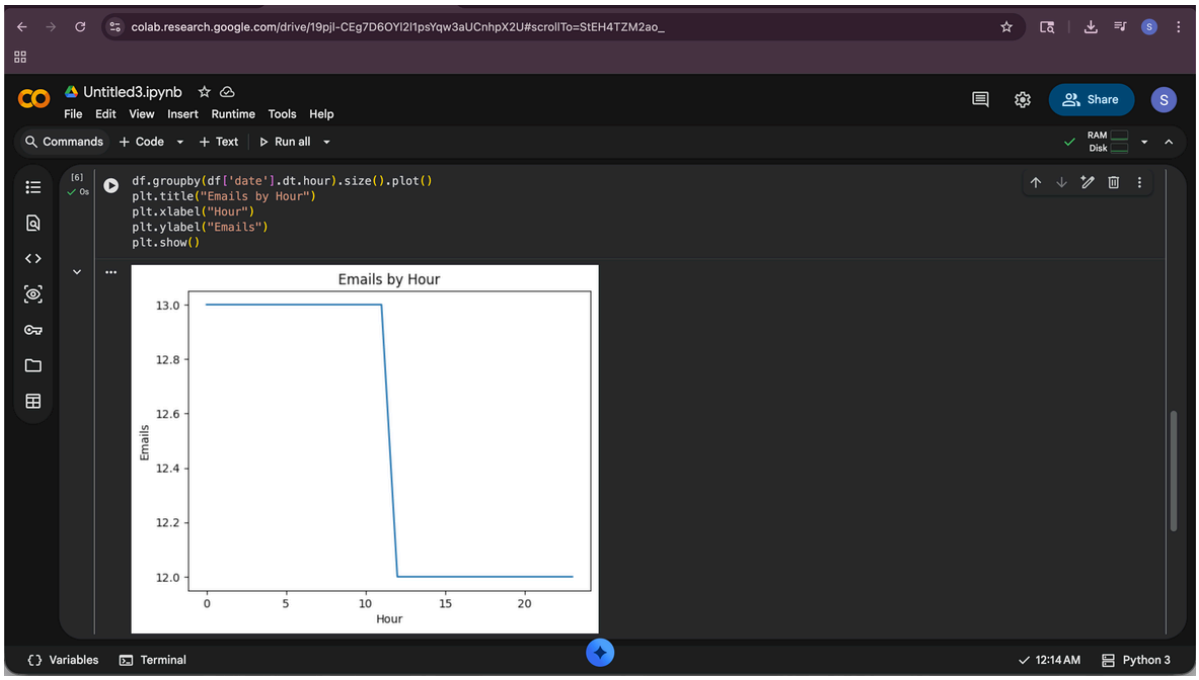
6. Graph Analysis with Theory

6.1 Bar Graph – Emails per Sender



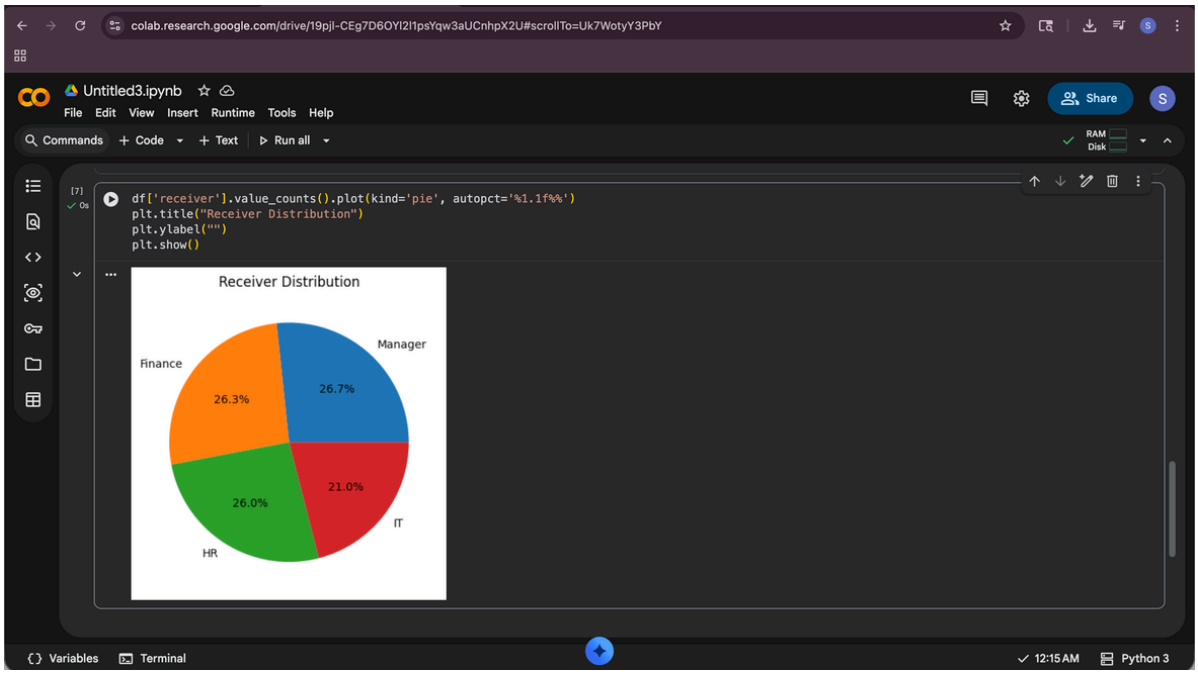
A bar graph compares categorical data. it helps identify the most active sender.

6.2 Line Graph – Emails by Hour



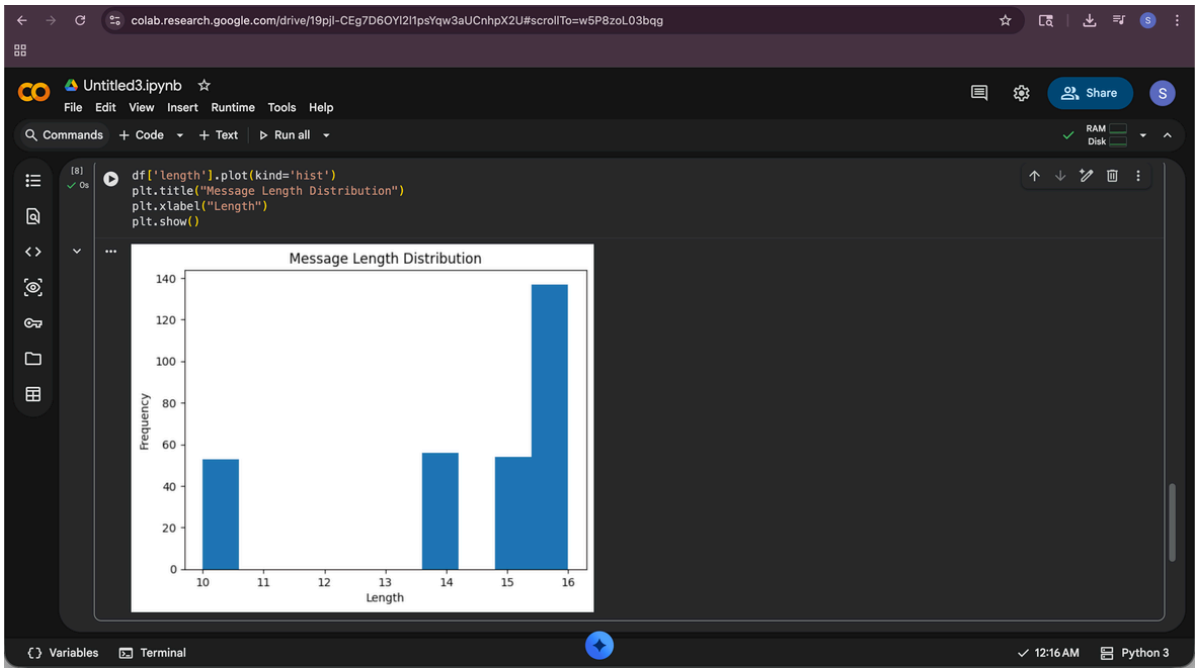
A line graph shows trends over time.

6.3 Pie Chart – Receiver Distribution



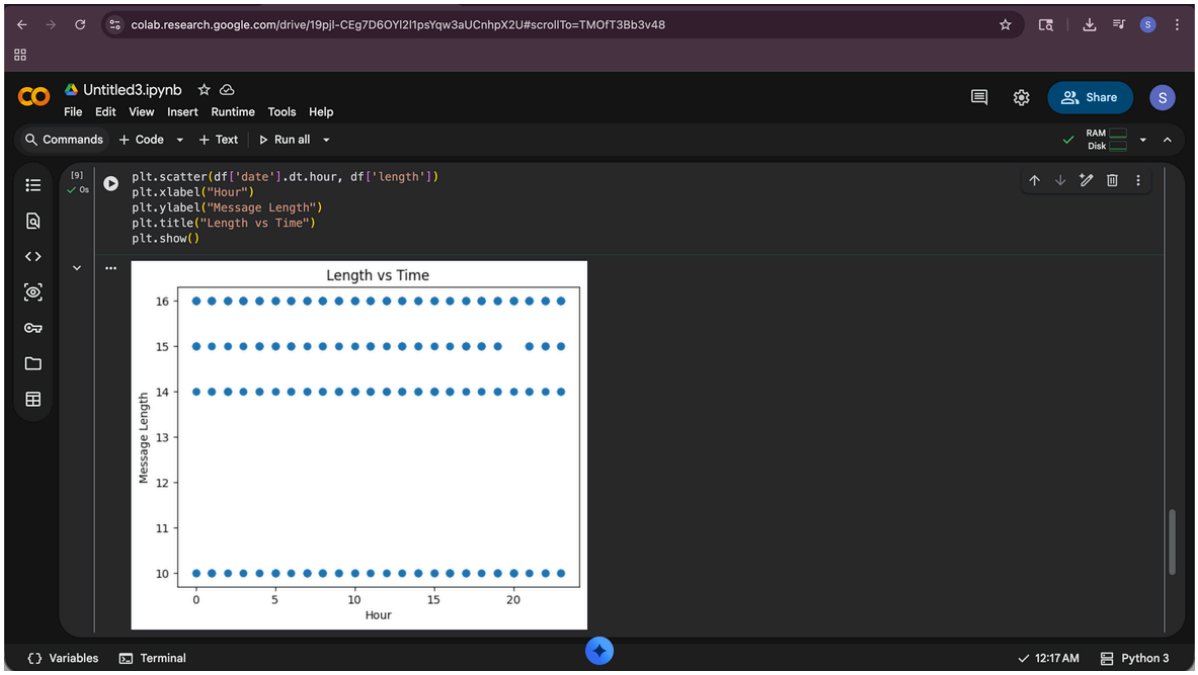
Pie chart shows proportions.

6.4 Histogram – Message Length



Histogram shows distribution of data.

6.5 Scatter Plot – Length vs Time



Shows relationship between two variables.

7. Key Findings

- Some senders are more active than others
- Emails are mostly sent during working hours
- Message lengths are generally short
- Communication varies across departments

8. Conclusion

The Enron Email Dataset provides valuable insights into organizational communication patterns. Using NumPy and Pandas, we can efficiently analyze and visualize large datasets.

Graphs help in:

- Understanding trends
- Identifying patterns
- Making better decisions